Oliver Cheung
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Summary

Postgraduate MEng Computer Science student with High-Performance Graphics and Games Engineering. Specialised in real-time rendering, parallel computation and game engine tools development, programmed in C++. Driven by an interest in games programming, seeking to apply strong technical foundations to the game industry, with a long term of goal of becoming a graphics programmer.

Projects

Boba Engine

- Collaborated with a team of 6 to produce a Vulkan Game Engine from scratch, with the goal of streamlining development for local co-op games
- Spearheaded Project Management, utilising Agile Software Development methods such as Kanban to dictate the team's goals, desired features, and final game design
- Played a key role in implementing the engine's mesh processing system to support static and animated meshes with the glTF format
- Includes MSAA, Toon Shading, Shadow Mapping and Ambient Occlusion to achieve a heavily stylised finish for the developer

OpenGL Boids Predator Model (BPM)

- Dissertation reapplying the Boids Algorithm against external predation
- Groups Boids together using DBSCAN to sort them into organised clusters, allowing the predator to target the largest flock
- Utilises a finite state machine to gauge the predator's current behaviour, alongside Boid escape patterns to represent the natural evolved reactions of prey

Elevending

- Tower defence game created in the Godot game engine, focused on reimplementing Pokémon's type-based combat
- Collaborated with members of other disciplines such as musicians, artists and film editor to ship the final product

Vulkan Renderer

- Real-time Vulkan renderer developed in C++
- Includes a debug pipeline to visualise texture mipmap level, fragment depth, and partial derivatives of said depth
- Utilises PBR shading for specular reflections, and a real-time mosaic effect to demonstrate the renderer's post processing capabilities

Mesh Processing Toolkit

- Built a mesh processing toolkit converting triangle soup meshes into a sorted face index format
- Implemented a directed edge mesh format to support manifold testing and mesh repair
- Applied Euler's formula to analyse a mesh's topology
- Utilised gaussian curvature for mesh simplification, whilst also preserving its manifold

OpenGL Raytracer

- CPU-side, Monte Carlo Raytracer implemented in OpenGL
- Applies Next Event Estimation for caustics and indirect lighting
- Includes Beer's Law and Fresnel Effect for transparency and reflections

OpenGL Animation / Physics Simulation

- Developed a real-time simulation featuring physically based rigid body estimation for sphere and dodecahedra
- Includes terrain-following skeletal animation for bipedal locomotion, with BVH driven interpolation
- Applies terrain surface normal data to compute rotational and linear impulse, leveraging mesh inertia tensors for accurate rotation calculation

Previous Games / Projects

- Titled Soul, Lift and Pilot, and were created in the Unity game engine
- Served as an introduction to the engine and improved proficiency in C#

Education

University of Leeds *Sep 2022 – July 2026*
MEng, Bsc Computer Science (High-Performance Graphics and Games Engineering)
Predicted: 1:1

The Charter School North Dulwich *Sep 2015 – June 2022*
A Levels: Mathematics (A), Further Mathematics (A*), EPQ (A*), Physics (B)*

Experience

Volunteer – Oxfam Headingley *Oct 2025 – June 2026*

- Delivered strong customer service in a fast-paced retail environment, maintaining clear communication through regular donor and customer interaction
- Meticulously organised, sorted and priced donated stock, ensuring accurate categorisation and consistent pricing standards
- Researched and identified valuable and vintage stock, enabling informed pricing decisions for donation revenue

Key Skills

Languages: C, C++, C#, GLSL, Python, SQL, Java, JavaScript, HTML, CSS

Technologies: Vulkan, OpenGL, Unity, Godot, Git, Premake, CMake, RenderDoc, gdb

Interests: Martial Arts, Cooking